

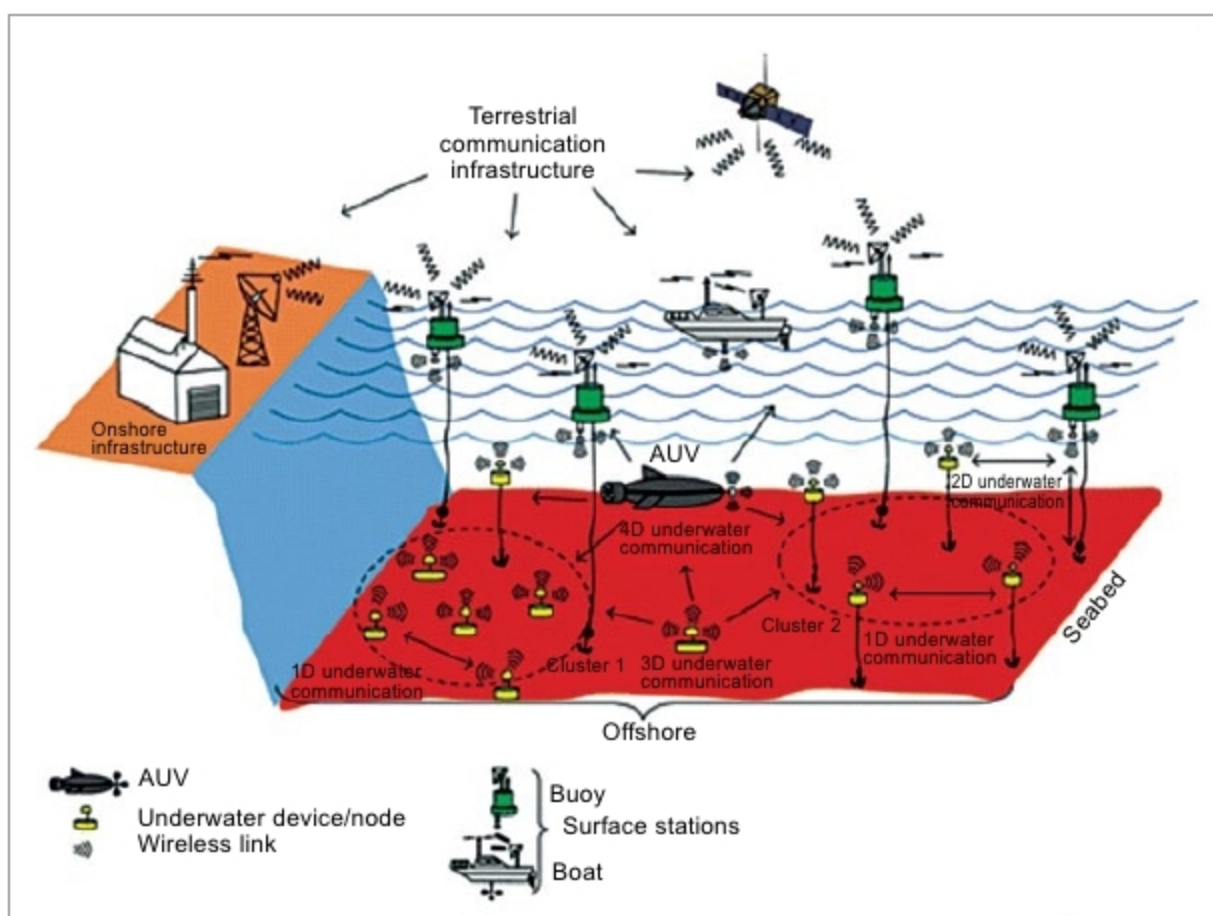


Fig. 8: A typical 4D UWSN architecture (Credit: www.researchgate.net)

has been developed. Schematic overview of the components of the high-altitude camera system for GEOMAR Remus 6000 AUV is shown in Fig. 7.

Underwater sensor network. Deep sea exploration requires a different approach for communication as compared to shallow water communication. Underwater applications include monitoring, disaster management, military, navigation and sports.

Underwater sensor network (UWSN) is used for underwater explorations. It is a network of autonomous sensor nodes that are spatially distributed underwater to collect temperature, pressure and other water-related data.

UWSN acoustic transceivers used for communication are low-frequency waves, but have long wavelengths suitable for long-distance communication. UWSN finds use in a wide range of applications for management and recovery of disaster monitoring and preventive mechanisms. UWSN with acoustic sensor networks and underwater mobile (ROVs, AUVs) sensor networks can be used on a large scale for exploration in the deep sea.

Designing for UWSN applications is based on common UWSN architectures. One-dimensional (1D) UWSN architecture refers to a network where sensor nodes are deployed auto-

mously and work on a stand-alone network. Each sensor node directly transmits information to a remote station.

In 2D UWSN, communication is carried in two ways. Each member of the sensor cluster communicates with its anchor node (with horizontal communication link) and then the anchor node communicates with the surface buoyant node (with vertical communication link).

In 3D UWSN architecture, sensors are deployed in the form of clusters are anchored at different depths and heights.

4D UWSN architecture is a combination of fixed 3D UWSN and mobile UWSN. Mobile underwater ROVs collect data from anchor nodes and relay it to the remote station. A typical 4D UWSN architecture is shown in Fig. 8.

Oceans of the future

Development for the next-generation ocean monitoring sensor system has already begun in the US, Japan and some European countries. It features a device called node, which is installed at the bottom of the sea and is connected to various sensor systems using an ROV.

US Defense Advanced Research Projects Agency (DARPA) has proposed intelligent floats similar to the

Some leading manufacturers of deep-sea sensors

- AMT Analysenmesstechnik GmbH
- CODAR Ocean Sensors
- Edaphic Scientific
- GE Sensing
- Honeywell
- Ocean Sensor Systems
- Omni Instruments
- Sea-Bird Scientific
- Subsea Sensors Inc.
- Systems Engineering & Assessment Ltd

Internet of Things (IoT). Instead of linking devices and sensors to the Internet, the Ocean of Things wire the world's oceans with a network of sensors, capable of recording vast amounts of data about the environment.

US Ocean Observatories Initiative provides researchers with spatial and temporal data from the ocean's surface all the way down to the deep sea. It is planning to launch robots to chart the world's oceans, with the goal of mapping everything by 2030.

A new cable network infrastructure has been developed for future expansion to meet the growing needs of science and technology over the next two to three decades (lifespan) of the network. Node functionality and sensor arrays in the network include seismic array, full water column mooring and tsunami warning system.

To sum up

Various environmental conditions of the sea, including temperature, pressure and currents, can be easily monitored with sensors. Modern electronic sensors, microelectronics, nanotechnology, powerful computers and high-bandwidth cables in the oceans will likely lead to unprecedented advancements in ocean exploration and sensing in the next few decades.

The deep sea needs to be explored not only for extracting rich resources for human needs, but also for detecting earthquakes and other related natural disasters for human safety. Underwater sensor networks along with ROVs and AUVs will help humans explore deeper in the sea. **EFY**